Gestão de Projecto de Software

Licenciatura em Engenharia Informática

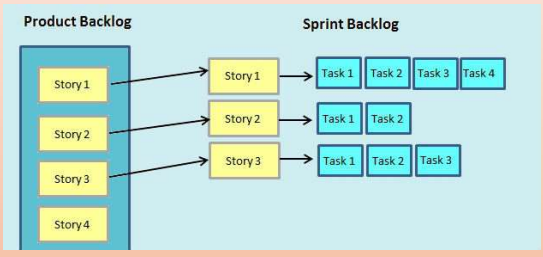
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SCRUM (2)

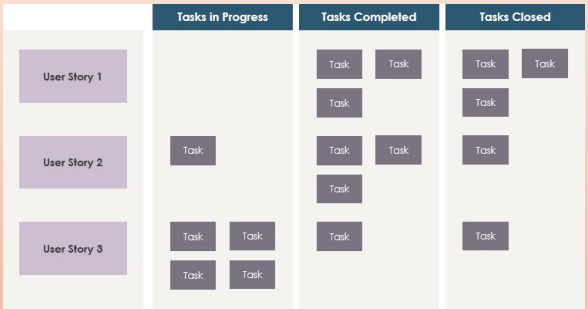
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Product and Sprint Backlog



Sprint Backlog – Scrum Board

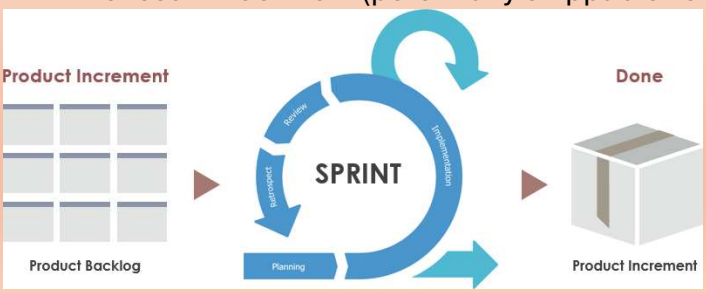


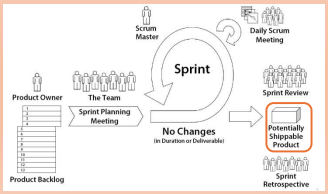
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Product Increment

- The (working) result at the end of each sprint
- The sum of all the product backlog items completed during a sprint plus the value of the increments of all previous sprints
- The result must work (potentially shippable/releasable) – no presentations.







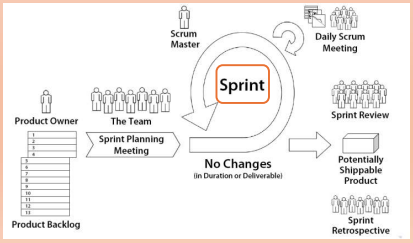
Events

Scrum

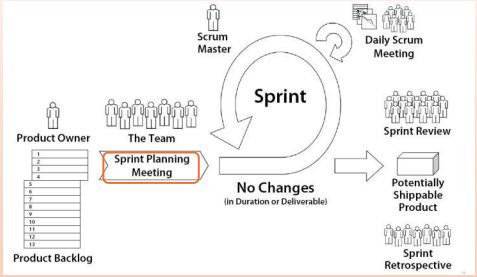


Sprint

- Fixed period iteration
  - typically between 2- and 4-weeks
- Includes design, coding, testing, and documentation
- Created "done" work – According to acceptance criteria
- During the Sprint, the Scrum Team keeps committed to the goals and end-date
- The Scrum Team is fully responsible for the work
  - Only the Scrum Team can add or remove items in the Sprint backlog
- Abnormal termination of Sprint may be called by the PO when the Sprint Goal no longer makes sense
- Sprints occur one after another, without any down-time between them.



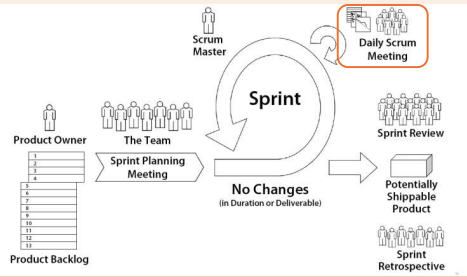
Sprint Planning Meeting



- 2-part meeting to define the next Sprint
- Sprint prioritization
  - Product Owner leads, all Stakeholders are present
  - Analyze and evaluate product backlog
  - Select Sprint goal

- Sprint planning
  - Scrum Team and Product Owner (optional)
  - Decide how to achieve sprint goal (design)
  - Create Sprint Backlog (tasks) from Product Backlog (features/user stories)
  - Answer
    - Why is this Sprint valuable?
    - What can be Done this Sprint?
    - How will the chosen work get done?
  - Estimate Sprint Backlog

Daily Scrum Meeting

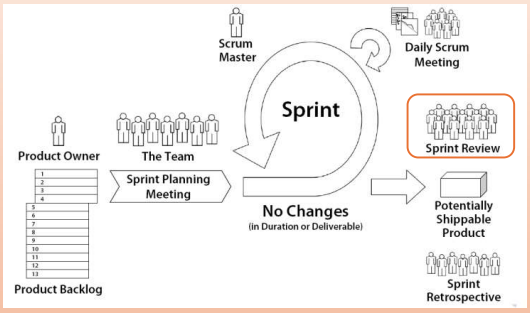


- Short meeting to update each other on progress and problems
  - Daily, stand up, 15 minutes
  - Usually same place, same time, at the beginning of the day

- Everyone answers 3 questions (not necessarily):
  - What have you done since last daily Scrum?
  - What will you do between now and the next daily Scrum?
  - What is getting in the way (blocks) of meeting the iteration goals?
- Scrum Master notes blocks, and afterwards helps resolve them
  - Not for problem solving
  - Other follow-up meetings can be held
- Others (chickens) can attend the meeting but they do not speak.

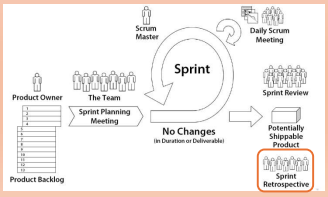
Sprint Review Meeting

- At the end of the Sprint.
- Product Owner, Team, Scrum Master, and all Stakeholders.
- Demo of what the team has produced.
- The Product Owner gathers feedback from everyone on ways to improve what's been built
  - This feedback is incorporated into the Product Backlog
- **Max 4 hours.**
- Goal: Inspect product increment.



Sprint Retrospective Meeting

- Also at the end of the Sprint
- The Team, Product Owner, and Scrum Master
- Goal: review the way of working and look for ways to improve their effectiveness.
- Mechanism for continuous improvement:
  - What went wrong, what went right, what can be done better
  - Which tools, processes, practices can be used to make it better, more productive and enjoyable
- Might be anonymous



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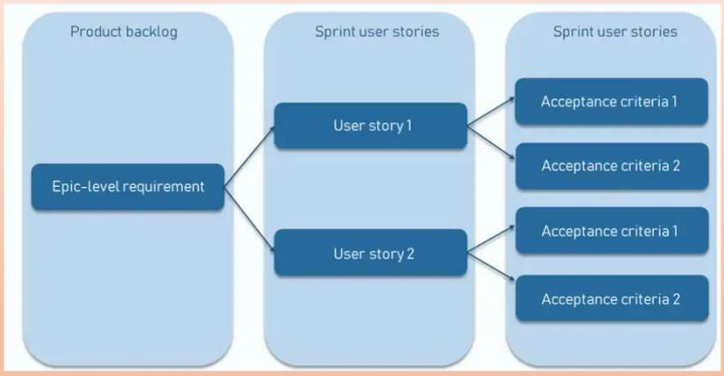
Scrum Values



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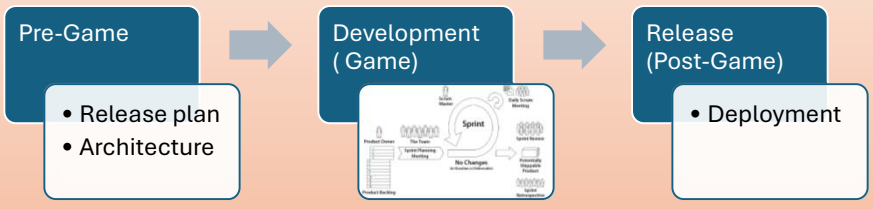
Product Backlog // User Stories / Sprint // Acceptance criteria



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Scrum Lifecycle



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Lifecycle

1. Pre-game
   1. Planning
      - Write vision, budget, User Stories, Product Backlog with estimates, exploratory design and prototypes
   2. Staging
      - Define requirements for next release, plan next release, exploratory design and prototypes
2. Development
   - Sprint planning, Sprint, Sprint review and retrospective
3. Release (Post-game)
   - Deployment, documentation, training, …

Further Reading

- How to write a Good User Story https://www.youtube.com/watch?v=P3hbbKnzm4A
- https://www.atlassian.com/agile/project-management/user-stories
- https://www.agilealliance.org/glossary/user-stories/
- Alexsoft, Acceptance Criteria for User Stories: Purposes, Formats, Examples, and Best Practices, 18/5/2021 (https://www.altexsoft.com/blog/business/acceptance-criteria-purposes-formats-and-best-practices/)

Bibliography

- http://www.scrumalliance.org
- https://www.scrum.org/
- https://scrumguides.org/scrum-guide.html

  - Some figures from the slides were taken from these links

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Risk Management

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Risk Plan

- A risk plan is a list of all risks that threaten the project, along with a plan to mitigate some or all of those risks.
  - The project manager selects team members to participate in a risk planning session:
    - The team members brainstorm potential risks
    - The probability and impact of each risk is estimated
    - A risk plan is constructed

RISK ?

- Project Risk is an uncertain event or condition that, if it occurs, has a positive or negative effect on, at least, one project objective, such as time, cost, scope or quality [PMBOK Guide 2004].
- **RISK is the possibility of suffering loss**

Conceito transversal a váreas áreas

- Gestão do risco
  - Conjunto de actividades que visam avaliar de uma forma cíclica a exposição de um recurso ao dano (risco) e que apontam soluções para reduzir esse risco



Risk management

- Risk management is concerned with identifying risks and drawing up plans to minimise their effect on a project.
- A risk is a probability that some adverse circumstance will occur
  - **Project risks** affect schedule or resources;
  - **Product risks** affect the quality or performance of the software being developed;
  - **Business risks** affect the organisation developing or procuring the software.

The risk management process

1. **Risk identification**
   - Identify project, product and business risks;
2. **Risk analysis**
   - Assess the likelihood and consequences of these risks;
3. **Risk planning**
   - Draw up plans to avoid or minimise the effects of the risk;
4. **Risk monitoring**
   - Monitor the risks throughout the project;

1. **Risk identification**
2. Risk analysis
3. Risk planning
4. Risk monitoring



Risk Management

1 - Risk identification

1. Risk identification

- Team activity
- A checklist of common risks may be used to identify risks in a project
  - Technology risks.
  - People risks.
  - Organisational risks.
  - Requirements risks.
  - Estimation risks.

Threshold of Success (ToS)

- The minimum set of conditions that must be met for the project to be considered successful
- If the team fails to meet **even one** of the conditions, then the project is a **failure**
- Just define 2-3 **SMART** goals.
  - **S**hort/**S**pecific, **M**easurable, **A**chievable, **R**elevant, and **T**ime bound.

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How to create a ToS

- First create a picture of failure, then convert failure into success.

Ex:

- The project fails if:
  - Essential features are not ready by the first release date.
  - Team members are dissatisfied or bored with their jobs.
- The project is successful if:
  - By the first release date, all "Must Have" features are implemented and pass acceptance tests with no known critical defects.
  - All team members give average score of 5 or better on a job satisfaction survey taken quarterly.

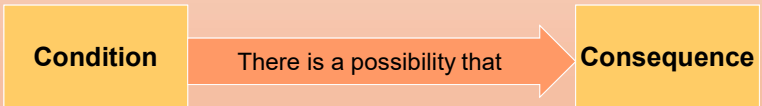
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Risk Statement

- Risks are captured in statements to:
  - Help understand those issues/concerns that might stop the project from being successful, i.e, fail to meet the ToS.
  - Provide consistency in how the information is displayed within and across projects.
  - Understand and follow project concerns.
  - Better communicate concerns to project stakeholders
- Statements create a concise description of particular issues or concerns that can be understood and acted upon.

Guidelines

- Consider and capture those **fact-based conditions (true now!)** that are causing concern for a potential loss to the project;
- followed by a brief description of the **potential negative consequences** of these conditions.



What does a good statement look like?

- **Condition**: a single phrase or sentence that briefly describes the key circumstances, situations, etc., that have caused concern, doubt, anxiety or uncertainty.
- **Consequence**: a single phrase or sentence that describes the key negative outcome(s) that might result.
- Example:
  - **Tight schedules are causing people to work too much overtime**; (there is a possibility that this) **might cause project burnout and lower team morale**.

Examples of different risk types

Risk type	Possible risks: condition;	consequence
Technology	The database used in the system cannot process as many transactions per second as expected;	performance requirements might not be met
People	Only one team member is expert in swing, and there are many screens to develop;	might delay GUI development, delaying the whole project.
Organizational	Organization has financial problems;	might force reductions in the project budget.
Tools	The code generated by software code generation tools is inefficient;	might compromise software quality.
Requirements	Customers fail to understand the impact of requirements changes;	requested changes might require major design rework .
Estimation	The team has no experience in software estimation;	might case underestimation, delaying the project in an uncontrolled way.

1. Risk identification
2. **Risk analysis**
3. Risk planning
4. Risk monitoring



Risk Management

2 - Risk analysis

2. Risk analysis

- Assess probability and impact of each risk.
- Risk ***impact*** might be catastrophic(5), serious(3), or tolerable(1).
- ***Probability*** may be very low(1), low, moderate, high or very high(5).
- Prioritize risks according to ***probability x impact***

Risk matrix

P x I		Impact				
		Tolerable (1)	(2)	Serious (3)	(4)	Catastrophic (5)
Probability	Very low (1)	Low (1)	Low (2)	Low (3)	Low (4)	Low (5)
	Low (2)	Low (2)	Low (4)	Low (6)	Low (8)	Medium (10)
	Moderate (3)	Low (3)	Low (6)	Low (9)	Medium (12)	Medium (15)
	High (4)	Low (4)	Low (4)	Medium (12)	Medium (16)	High (20)
	Very high (5)	Low (5)	Medium (10)	Medium (15)	High (20)	High (25)

1. Risk identification
2. Risk analysis
3. **Risk planning**
4. Risk monitoring



Risk Management

3 - Risk planning

3. Risk planning

- Consider the top risks (e.g. with probability x impact >= 20) and develop a strategy to mitigate those risks.

- **Avoidance strategies** (*Plano de Prevenção*)
  - The probability that the risk will arise is reduced;
- **Minimisation strategies** *(Plano de Minimização)*
  - The impact of the risk on the project or product will be reduced;
- **Contingency plans** *(Plano de Contingência)*
  - If the risk arises, contingency plans are to deal with that problem;



1. Risk identification
2. Risk analysis
3. Risk planning
4. **Risk monitoring**

Risk Management

4 - Risk monitoring

4. Risk monitoring

- Assess each identified risks **regularly** to decide whether or not it is becoming less or more probable.
- Also assess whether the effects of the risk have changed.
- Each key risk should be discussed at management progress meetings.

Further reading

- Ian Sommerville, *Software Engineering 9*, Pearson Education, 2011. [Section 22.1]

- Some slides adapted from Ian Sommerville, “Software Engineering”


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